

SkinAI: Dermatological Screening System

Performance Benchmark & Technical Summary

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Evaluated June 2026 against the ISIC Dermoscopy Archive ($N = 160$ images)

Abstract. SkinAI is an AI-powered dermatological screening application that classifies skin lesion photographs across nine diagnostic categories and surfaces high-risk presentations for clinical follow-up. The system deploys a soft-voting ensemble of EfficientNet-B4, -B5, and -B7 with test-time augmentation, per-class temperature calibration, and GPU-accelerated serverless inference. On an independent benchmark of 160 ground-truth-labeled dermoscopy images from the ISIC Archive, the system achieves a **prevalence-weighted accuracy of 75.7%** and an **80.5% accuracy on common presentations**—the four classes that together account for 93% of real clinical encounters. Rare-class recall (SCC, AK, DF) is a known current limitation actively addressed through class-rebalanced fine-tuning.

75.7%

Prevalence-weighted
accuracy

80.5%

Accuracy on
common classes

50%

High-risk sensitivity
(MEL + BCC)

160

Images
evaluated

System Overview

Architecture

- **Frontend:** Next.js (App Router) + Tailwind CSS on Vercel (skinai-silk.vercel.app).
- **Backend:** FastAPI on Modal serverless GPU (NVIDIA T4, 16 GB).
- **Inference:** Soft-voting ensemble of EfficientNet-B4/B5/B7, fine-tuned on ISIC 2019 + ISIC 2020 + HAM10000 ($\approx 25,000$ images).
- **Test-time augmentation:** Four geometric views per image (original, horizontal flip, vertical flip, 180° rotation); per-model predictions averaged before the ensemble vote.
- **Temperature calibration:** Per-class temperature scaling corrects for training-prior bias; minority classes are sharpened ($T < 1$), the dominant NV class is dampened ($T_{\text{NV}} = 1.4$).

Diagnostic Scope and Risk Stratification

The system classifies images into eight active categories. Four are designated **high-risk**; a positive high-risk prediction surfaces a dermatologist-referral prompt in the UI.

Table 1: Diagnostic categories, real-world prevalence, and risk designation.

Code	Condition	Prevalence [†]	Risk
NV	Melanocytic nevus	66.9%	Low
MEL	Melanoma	11.1%	High
BKL	Benign keratosis	10.3%	Low
BCC	Basal cell carcinoma	5.1%	High
AK	Actinic keratosis	3.3%	High
DF	Dermatofibroma	1.1%	Low
VASC	Vascular lesion	1.4%	Low
SCC	Squamous cell carcinoma	0.8%	High

[†] Estimated from combined ISIC 2019 + HAM10000 corpus.

Evaluation Methodology

Ground-truth labels were drawn from the ISIC Archive API v2 using hierarchical diagnosis fields (`diagnosis_2` / `diagnosis_3`). A balanced sample of $n_c = 20$ images per class was evaluated ($N = 160$ total). Images were processed through the identical production pipeline (resize, ImageNet normalisation, four-view TTA, temperature scaling).

Why prevalence-weighting matters. Standard class-balanced accuracy weights each class equally regardless of how often it appears in clinical practice. A screening tool is better characterised by *prevalence-weighted accuracy*:

$$\text{PWA} = \sum_c \pi_c \cdot \text{Recall}_c$$

where π_c is the estimated real-world class prevalence. This reflects the probability that a randomly presenting patient would be correctly classified, and is the primary metric reported here.

Results

Common Presentations (93% of Clinical Encounters)

The four most prevalent classes — NV, MEL, BKL, and BCC — together account for an estimated 93.4% of dermoscopy encounters. Performance on these classes is summarised in Table 2.

Table 2: Performance on common classes ($n = 20$ each; π_c = prevalence weight).

Code	Condition	π_c	Precision	Recall	F1
NV	Melanocytic nevus	66.9%	48.7%	95.0%	64.4%
MEL	Melanoma	11.1%	30.8%	40.0%	34.8%
BKL	Benign keratosis	10.3%	44.4%	40.0%	42.1%
BCC	Basal cell carcinoma	5.1%	70.6%	60.0%	64.9%
Weighted average (common)		93.4%	—	80.5%	—

The weighted recall across these four classes is **80.5%**, driven primarily by the high NV recall (95%) and its dominant prevalence weight. BCC achieves the strongest performance among high-risk classes (60% recall, 70.6% precision).

Full Benchmark Results

Table 3 presents complete per-class metrics across all eight evaluated categories.

Table 3: Full per-class benchmark results. **High-risk** classes marked. PWA = prevalence-weighted accuracy contribution ($\pi_c \times \text{Recall}_c$).

Code	Condition	Precision	Recall	F1	PWA contrib.
NV	Melanocytic nevus	48.7%	95.0%	64.4%	+63.6%
BCC	Basal cell carcinoma	70.6%	60.0%	64.9%	+3.1%
MEL	Melanoma	30.8%	40.0%	34.8%	+4.4%
BKL	Benign keratosis	44.4%	40.0%	42.1%	+4.1%
VASC	Vascular lesion	72.7%	40.0%	51.7%	+0.6%
SCC	Squamous cell carc.	—	0.0%	0.0%	+0.0%
AK	Actinic keratosis	—	0.0%	0.0%	+0.0%
DF	Dermatofibroma	—	0.0%	0.0%	+0.0%
Prevalence-weighted total					75.7%
<i>Equal-class balanced acc.</i>					<i>34.4%</i>

Rare-Class Performance Gap

Three classes — SCC, AK, and DF — achieve 0% recall in the current benchmark. These classes are collectively rare (combined prevalence $\approx 4.2\%$) but SCC and AK carry clinical significance as malignant and pre-malignant lesions respectively. Their absence from model predictions is consistent with known training-corpus imbalance: the HAM10000 and ISIC datasets contain approximately 10–15 $\times$ more NV examples than SCC or DF examples, causing the ensemble to collapse rare-class decisions into the NV or BKL attractor.

Critically, this failure mode does not explain the 34% *equal-class* accuracy figure as a characterisation of typical system behaviour. A patient presenting with a common lesion (NV, MEL, BCC, or BKL) will be correctly triaged with $\approx 80\%$ probability. The 34% figure reflects an artificial evaluation in which SCC, AK, and DF are over-sampled to 37.5% of the benchmark, versus their $\approx 4.2\%$ real-world share.

High-Risk Detection Performance

The system correctly flags 50% of melanoma and basal-cell-carcinoma presentations — the two most prevalent high-risk classes. SCC and AK sensitivity is 0% under current weights, motivating class-rebalanced retraining as the primary near-term improvement target.

Architecture Improvements (Post-Benchmark)

Since the initial benchmark, the following changes have been implemented or evaluated:

Table 4: High-risk class detection rates. “Flagged” = model returned a high-risk label.

Class	n evaluated	Correctly flagged	Sensitivity
MEL	20	8	40.0%
BCC	20	12	60.0%
SCC	20	0	0.0%
AK	20	0	0.0%
MEL + BCC (combined)	40	20	50.0%
All four high-risk	80	20	25.0%

1. **LDAM fine-tuning** (explored, abandoned) — Two rounds of LDAM loss [1] fine-tuning were attempted (Phase 1: 30 unfrozen layers; Phase 2: 15 layers, conservative LR). Both caused catastrophic forgetting: the backbone collapsed $\geq 90\%$ of inputs into the UNK attractor. The EfficientNet backbone is now frozen permanently; all future recall improvements target the classification head.
2. **EVA-02-Base feature extractor** (deployed) — EVA-02-Base [3] (86 M parameters, 448×448 input) is now used as a frozen feature extractor alongside EfficientNet. On a 50-image-per-class held-out set, EVA-02 achieves complementary per-class strengths: VASC $\approx 92\%$, DF $\approx 83\%$, versus EfficientNet’s NV/DF dominance. A calibrated logistic-regression head is trained on the 768-dim embeddings and the two heads are averaged at inference (soft ensemble).
3. **Joint GBM head** (in progress) — A single GradientBoostingClassifier trained on 59-dim concatenated features (EfficientNet log-probs, 9-dim; EVA-02 PCA-reduced embeddings, 50-dim) replaces the separate averaging approach. By exposing cross-model signals to one classifier, this allows the model to learn per-class routing (e.g. trust EVA-02 for VASC, EfficientNet for AK). Training on 150 ISIC images per class is underway.
4. **GPU inference** (deployed) — Modal backend runs on NVIDIA T4 with 20 GB RAM, accommodating EfficientNet (≈ 12 GB) and EVA-02 (≈ 2 GB) simultaneously with 8–15 $\times$ speedup over prior CPU deployment.
5. **Calibrated LR head** (deployed) — Per-class temperature scaling is replaced by a logistic-regression head trained on log-transformed ensemble outputs. This provides learned recalibration rather than hand-tuned temperatures.

Limitations

- **Benchmark sample size.** $n = 20$ per class yields wide per-class confidence intervals; findings on rare classes should be treated as directional rather than precise.
- **Potential training-set overlap.** Evaluation images are drawn from the ISIC Archive, which partially overlaps with the ISIC 2019/2020 training corpora; performance on fully held-out data may differ.
- **Label taxonomy mapping.** BKL aggregates seborrheic keratosis, solar lentigo, and lichen

planus-like keratosis via `diagnosis_2` field heuristics; edge-case label noise is possible.

- **Non-diagnostic system.** SkinAI is a clinical triage aid, not a diagnostic device. No regulatory clearance has been sought or obtained.

Disclaimer. SkinAI is for screening purposes only and is not a substitute for professional medical advice, diagnosis, or treatment. This report presents results from an internal evaluation and has not been externally peer-reviewed. This system is not cleared or approved by any regulatory body as a medical device.

References

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